Milagro: Predicting Store Profitability at a Fast-Casual Restaurant Chain

1 Dataset Preparation and Rationale

Kathleen's team has already split the Milagro store data into training and testing sets.

Questions

1. **Why did Kathleen's team split the data into a training set (374 stores) and test set (85 stores)?**
   (a) What percentage of the total data is in the training set?
   (b) Explain what the training set will be used for, what the test set will be used for, and why it is important not to use the test set during model building.

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2 Kathleen's Original Model

Kathleen originally built a multiple linear regression model using the training dataset to predict annual store profitability (`annual.profit`) as a function of four variables: `agg.inc`, `sqft`, `col.grad`, and `com60`.

Questions

2. **Fit a linear regression model using the training data with the four variables: `agg.inc`, `sqft`, `col.grad`, and `com60`.**
   (a) Write the complete linear regression equation for predicting annual store profitability from these four predictors. Your equation should be in the form:

$$\text{annual.profit} = \beta_0 + \beta_1 \times \text{agg.inc} + \beta_2 \times \text{sqft} + \beta_3 \times \text{col.grad} + \beta_4 \times \text{com60}$$

3. **Using the estimated regression model, what annual profitability is predicted for a Milagro store located in an area with:**

   - Aggregate income (`agg.inc`) of $100,000,000
   - Store size (`sqft`) of 800 square feet
   - College graduate percentage (`col.grad`) of 0.30 (30%)
   - Long commute percentage (`com60`) of 0.10 (10%)

4. **Evaluate the quality of the original model:**

   (a) What is the R^2 value on the training data?

   (b) What is the R^2 value on the test data?

5. **Test the statistical significance of the predictors:**

   (a) Which independent variables are statistically significant at the 5% level ($\alpha = 0.05$)?

   (b) Which variable has the smallest p-value (most statistically significant)?

   (c) Which variable has the largest p-value (least statistically significant, but still below 0.05)?

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3 Exploratory Correlation Analysis

Kathleen wants to understand the relationships between variables in the expanded dataset before building more complex models.

Questions

6. **Compute the correlation matrix for all numerical predictor variables** (exclude `store.number`, `annual.profit`, and `state`).

   (a) The dataset now has 10 predictor variables: the 4 original variables plus 6 new variables. Identify the **three pairs** of variables with the strongest correlations (highest absolute values). Report the correlation coefficient for each pair.

7. **Statistical significance of new variables:** Build a regression model using ALL 10 variables (the 4 original plus 6 new variables). Test the statistical significance of each variable at the 5% level ($\alpha = 0.05$).

(a) Which of the 6 new variables (`lci`, `nearcomp`, `nearmil`, `freestand`, `gini`, `housemed`) are statistically significant?

(b) Which of the new variables are NOT significant? What does this suggest about their usefulness in predicting store profitability?

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4 Model Comparison

Now build and compare four different models.

Questions

8. **Fit and evaluate four models using the training data:**

Model A: Kathleen's Original Model

Variables: `agg.inc`, `sqft`, `col.grad`, `com60`

Model B: Full Model

Variables: All variables except `store.number`, `annual.profit`, and `state`

Model C: Parsimonious Model

- Build this model by removing variables that meet either of these criteria:
  - Variables that are NOT statistically significant at the 5% level (from Question 7).
  - Variables involved in pairs with absolute correlation > 0.70 (from Question 6). For highly correlated pairs, keep the variable with stronger correlation to the outcome variable (`annual.profit`).

Model D: Alternative Model

- Start with the original 4 variables (`agg.inc`, `sqft`, `col.grad`, `com60`).
- Add ONE variable from the 4 significant new variables identified in Question 7: `lci`, `nearcomp`, `nearmil`, `freestand`.

- Test each of the 4 possible additions (one at a time) and choose the one that:
  - Improves test R^2 compared to Model A, **and**
  - Maintains total profitability prediction $\geq$ \$40 million.

(a) For Model D report which variable you added.

(b) For each model, report:

i. Training R^2

ii. Test R^2

iii. Total predicted profitability for the 48 construction sites (in millions)

9. **Model recommendation and the dilemma:** Review the performance of your four models. You should notice a critical dilemma: Models with the highest test R^2 (best predictive accuracy) predict profitability BELOW \$40 million, while models that meet the \$40 million target have lower test R^2.

   - Which model would you recommend to Harriman Capital? In your answer, discuss whether you should prioritize statistical performance (higher test R^2) even if it means revising the \$40M profitability estimate downward, or prioritize meeting the business requirement (\$40M target) even with lower predictive accuracy. What are the business risks of each choice?